

Muhammad Umar Khan

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EDUCATION

Wayne State University | Detroit, MI, USA | Jan. 2025 - Dec. 2026

M.S. in Computer Science · GPA: 3.89/4

University of Windsor | Windsor, ON, Canada | Sep. 2020 – Apr. 2024

B.S. in Computer Science – Software Engineering Specialization

TECHNICAL SKILLS

Languages: Java, Python, JavaScript/TypeScript, HTML, CSS, SQL, C

Frameworks & Libraries: Spring Boot, Spring Security, Hibernate/JPA, React, Next.js, Node.js, Vite, Tailwind CSS, Recharts, Flask, Ollama

Databases & Tools/DevOps: PostgreSQL, SQLite, MongoDB, Supabase, Git, GitHub, Docker, REST APIs, JWT

Data & AI/ML: NumPy, Pandas, Matplotlib, TensorFlow, PyTorch, Scikit-learn, CNN+LSTM, RAG, TF-IDF, Time Series Forecasting

PROJECTS

MindForge AI Suite (StudyAI + ResearchAI) | Python, Flask, Ollama, LLaMA 3.2, RAG, TF-IDF, ThreadPoolExecutor

- Built a local study and research suite powered by LLaMA 3.2 via Ollama, running 100% offline with support for PDF, DOCX, PPTX, and TXT formats.
- Implemented an NLP retrieval pipeline using RAG with TF-IDF chunk scoring, reducing context window usage and improving response relevance across multi-document research sessions.
- Optimized multi-document analysis throughput by ~75% using parallel summarization via ThreadPoolExecutor, with real-time SSE streaming for live progress updates.
- Built a cross-document NLP synthesis engine with web search integration (DuckDuckGo), APA citation generation, and structured research report generation via LLM.

Market Predictor | React, Next.js, Supabase, Claude AI, Recharts, JavaScript

- Implemented a custom CNN+LSTM neural network from scratch in JavaScript for stock & crypto price forecasting, training live over 30 epochs with real-time loss curve visualization across 1W/2W/1M horizons.
- Integrated Claude AI for chart image analysis (OCR + pattern recognition), investment insights, and news sentiment — routed through a secure Next.js server-side API proxy.
- Built a prediction accuracy tracker that resolves forecasts against real market prices post-horizon and benchmarks CNN+LSTM against flat and momentum baselines.

Blog Platform (Full-Stack) | Java, Spring Boot, Spring Security, JWT, React, TypeScript, Vite, Tailwind CSS, PostgreSQL

- Built a secure full-stack blog platform with JWT authentication and role-based access control using Spring Boot and Spring Security, and architected RESTful APIs with layered architecture (Controller → Service → Repository).
- Developed a responsive React + TypeScript frontend with CRUD operations, dynamic filtering, secure password hashing, token refresh flow, and protected route handling for session management.

K-Means Market Basket Analysis | Java

- Built data-driven predictive models including K-Means clustering and the Apriori algorithm from scratch on a ~25% subset of a 530k+ entry dataset to analyze customer purchasing patterns, while generating association rules and cluster insights for data-driven decision making.

EXPERIENCE

Research Assistant

University of Windsor | Windsor, ON, Canada | Apr. 2023 - Nov. 2024

- Engineered a Python-based image segmentation pipeline using Euclidean-distance clustering to generate super pixels, reducing preprocessing time by approximately 20%.
- Evaluated CNN and Graph-CNN architectures for pixel classification, benchmarking accuracy and runtime to recommend the highest-performing model for research publication.
- Modeled and integrated an SQL database with Snowflake and containerized prototypes using Docker.

Teaching Assistant

University of Windsor | Windsor, ON, Canada | Sep. 2022 – Apr. 2024

- Conducted weekly office hours and graded assignments/exams for 100+ students enrolled in Data Structures & Algorithms, Database Management Systems, and Object-Oriented Programming (Java).
- Assisted lab sessions and exams, and aided in reinforcing core concepts in algorithms, database systems, and OOP.